Dr Bimandra Djaafara (Andra)

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Senior Research Fellow

SUMMARY

I am an infectious disease epidemiologist and mathematical modeller with over a decade of experience using data and models to support infectious disease control programmes in Indonesia and other low- and middle-income countries. I hold a PhD from Imperial College London, specialising in translating research into practical recommendations for public health policy. My work encompasses COVID-19 response, malaria and dengue control, measles elimination modelling, and outbreak responses, utilising diverse data sources, including routine surveillance data, population surveys, and call detail records (mobile phone data). I have collaborated extensively with Indonesia's Ministry of Health and local government health departments, trained over 300 public health officers, secured competitive research funding including the MORU-OUCRU Discovery Research Academy Fellowship, and published research in leading medical and public health journals. Currently based at the National University of Singapore, I am committed to bridging academic research and real-world public health practice through innovative modelling approaches and sustained capacity building in Indonesia and across low- and middle-income countries.

EDUCATION

- **PhD in Epidemiology, Evolution, and Control of Infectious Diseases**, Imperial College London 2019 - 2023
- **MRes in Biomedical Research (with distinction)**, Imperial College London 2018 - 2019
- **MSc in Epidemiology (with distinction)**, Imperial College London 2015 - 2016
- **BSc in Statistics**, IPB University 2008 - 2012

EMPLOYMENT

- **Senior Research Fellow**, Saw Swee Hock School of Public Health, National University of Singapore 2024 - Present
- **Research Associate**, MRC Centre for Global Infectious Disease Analysis, Imperial College London 2023 - 2024
- **Statistician & Mathematical Modeller**, Eijkman-Oxford Clinical Research Unit (now: OUCRU Indonesia) 2014 - 2018

SELECTED PUBLICATIONS

Summary: 23 peer-reviewed publications | 7 first author | Citations: 2157 | h-index: 13

First Author

1. **Djaafara BA**, Elyazar IRF, Silalahi FSM, Surya A, …, Mishra S, et al. (2026). Dengue transmission heterogeneity across Indonesia's archipelago: Climate-driven spatiotemporal patterns and policy implications. *PLOS Neglected Tropical Diseases*. link
2. **Djaafara BA**, Adrian V, Eriawati E, Elyazar IRF, Hamers RL, Baird JK, Thwaites GE, Clapham HE (2026). Modeling the transmission dynamics and control strategies during the 2017 diphtheria outbreak in Jakarta, Indonesia. *Infectious Disease Modelling*. link
3. **Djaafara BA**, Elyazar IRF, Yosephine P, Surya A, Silalahi FSM, …, Mishra S, et al. (2026). Bridging surveillance gaps in dengue: a hierarchical model integrating mixed data sources for transmission estimation and vaccine targeting. *medRxiv*. **[Preprint]** link
4. **Djaafara BA**, Elyazar IRF, Surya A, Silalahi FSM, …, Mishra S, et al. (2025). Estimating dengue force of infection from age-stratified surveillance data in Java, Indonesia. *Journal of The Royal Society Interface*. link
5. **Djaafara BA**, Mutunga M, Eneanya OA, Forna A, Cucunuba ZM (2025). The bridge between two worlds: Global South researchers' journeys through Global North academic training and beyond. *Epidemics*. link
6. **Djaafara BA**, Sherrard-Smith E, Churcher TS, Fajariyani SB, Prameswari HD, Herdiana H, Puspadewi RT, Lestari KD, Elyazar IRF, Walker PGT (2025). Spatiotemporal heterogeneity in malaria transmission across Indonesia: analysis of routine surveillance data 2010-2019. *BMC Medicine*. link
7. **Djaafara BA**, Whittaker C, Watson OJ, Verity R, Brazeau NF, et al. (2021). Using syndromic measures of mortality to capture the dynamics of COVID-19 in Java, Indonesia, in the context of vaccination rollout. *BMC Medicine*. link
8. **Djaafara BA**, Imai N, Hamblion E, Impouma B, Donnelly CA, Cori A (2021). A Quantitative Framework for Defining the End of an Infectious Disease Outbreak: Application to Ebola Virus Disease. *American Journal of Epidemiology*. link

Co-Author

1. Hadley L, Ben Miled S, Clapham HE, **Djaafara BA**, et al. (2026). How understanding the diversity of perspectives and systems in governments can increase the impact of scientific research. *Nature Communications Health*. link
2. Surendra H, **Djaafara BA**, Prameswari HD, Supriyanto D, Waluyo P, et al. (2024). Mitigating risks of malaria and other vector-borne diseases in the new capital city of Indonesia. *Nature Communications*. link
3. Johnson R, **Djaafara B**, Haw D, Doohan P, Forchini G, Pianella M, Ferguson N, Smith PC, Hauck KD (2023). The societal value of SARS-CoV-2 booster vaccination in Indonesia. *Vaccine*. link
4. Fadilah I, **Djaafara BA**, Lestari KD, Fajariyani SB, Sunandar E, et al. (2022). Quantifying spatial heterogeneity of malaria in the endemic Papua region of Indonesia: Analysis of epidemiological surveillance data. *The Lancet Regional Health - Southeast Asia*. link
5. Surendra H, Salama N, Lestari KD, Adrian V, Widyastuti, Oktavia D, Lina RN, **Djaafara BA**, et al. (2022). Pandemic inequity in a megacity: a multilevel analysis of individual, community and healthcare vulnerability risks for COVID-19 mortality in Jakarta, Indonesia. *BMJ Global Health*. link
6. Whittaker C, Walker PGT, Alhaffar M, Hamlet A, **Djaafara BA**, Ghani AC, Ferguson N, Dahab M, Checchi F, Watson OJ (2021). Under-reporting of deaths limits our understanding of true burden of covid-19. *BMJ*. link

7. Surendra H, Elyazar IRF, **Djaafara BA**, Ekawati LL, Saraswati K, et al. (2021). Clinical characteristics and mortality associated with COVID-19 in Jakarta, Indonesia: A hospital-based retrospective cohort study. *The Lancet Regional Health - Western Pacific*. link

GRANTS & AWARDS

- **Economic Burden and Care-Seeking Pathways for Malaria in Pregnancy in Papua, Indonesia**, Strengthening preparedness in the Asia-Pacific region using knowledge, learning & Engagement (SPARKLE) (2026)
- **Modelling SIA strategies for measles elimination in Indonesia**, Measles Analytics Hub, Imperial College London (2026)
- **MORU-OUCRU Discovery Research Academy (MODRA)**, MORU-OUCRU Discovery Research Academy (MODRA) (2025 - 2026)
- **SSHSPH Research Fellow Research Award**, Saw Swee Hock School of Public Health, National University of Singapore (2025/2026)
- **Early career researcher bursary**, Isaac Newton Institute, University of Cambridge (2024)
- **Wellcome Data Science Ideathon Bronze Prize**, Wellcome (2023)
- **Student Awards for Outstanding Achievement Medal**, Imperial College London (2022)
- **MRes + PhD Studentship**, School of Public Health, Imperial College London (2018)

LATEST PRESENTATIONS

- Serocatalytic models for infectious disease control in Indonesia: turning mixed data sources into actionable understanding. *[Online Keynote Speech] Symposium on Biomathematics (Symomath) 2026*, Sep 2026 **[Presentation]**
- Using mathematical models to inform dengue vaccination decisions. *Asia Pacific Vaccine Research Network (APVRN) Masterclass 2026*, Sep 2026 **[Presentation]**
- Bayesian Thinking and Analysis for Health Research. *Faculty of Medicine, Public Health and Nursing, Universitas Gadjah Mada*, Aug 2026 **[Presentation]**
- Bridging surveillance gaps in dengue: a hierarchical model integrating mixed data sources. *Indonesia-UK Initiative for Vector-Borne Disease Modelling Workshop*, Aug 2026 **[Presentation]**
- Bayesian hierarchical catalytic model to estimate dengue force of infection. *ESPIDAM Summer School 2026*, Jun 2026 **[Poster]**
- Modelling subnational measles SIA campaigns in Indonesia (preliminary results). *Measles Analytics Hub annual meeting 2026*, Jun 2026 **[Presentation]**
- Modelling for policy in Indonesia: Reflections from a decade of work with Indonesia's health programmes. *PREPARE - TF READI: Policy-driven Infectious Diseases Modelling Workshop*, May 2026 **[Presentation]**
- Modelling to inform policymaking. *Collaborative workshop on Infectious Disease Modelling for Health Policy and Research*, Oct 2025 **[Presentation]**
- Subnational dengue risk mapping of Indonesia. *Dengue multisource collaborative surveillance workshop (WHO Indonesia & Indonesia Ministry of Health)*, Apr 2025 **[Presentation]**
- Modelling to inform policymaking: experience from Indonesia. *Modelling and inference for pandemic preparedness workshop, Isaac Newton Institute*, Aug 2024 **[Presentation]**

TEACHING

- Fundamentals of Infectious Disease Modelling (Lecturer), NUS
- Mathematical modelling of infectious diseases: (why + how) we do it? (Guest Lecturer), Monash University Indonesia, University of Indonesia
- Introduction to Infectious Disease Modelling (Teaching Assistant), Imperial College London
- Introduction to Statistical Thinking and Data Analysis (Teaching Assistant), Imperial College London
- Mathematical Modelling of Infectious Diseases, Monash University Malaysia (SPARKLE)
- From Epidemiological Model to Public Health Decision: An Introduction to Infectious Disease Modelling for Public Health Practitioners, Center for Tropical Medicine, Universitas Gadjah Mada
- Modelling and Analysis of Serological Data: An Introductory Short Course, Imperial College London & OUCRU Vietnam
- Basics of Model Fitting, MIDSEA Summer School - Inference Track
- Infectious Disease Transmission Modelling for Mitigating Potential Epidemic Threats, 2nd Gadjah Mada International Conference of Tropical Medicine (GAMA - ICTM)

SKILLS

Programming: R, Stan | **Methods:** Bayesian inference, Infectious disease modelling, Statistical modelling | **Languages:** English, Indonesian

REFERENCES

Dr Swapnil Mishra (Saw Swee Hock School of Public Health, National University of Singapore) | **Dr Patrick Walker** (Department of Infectious Disease Epidemiology, Imperial College London) | **Dr Iqbal Elyazar** (Oxford University Clinical Research Unit (OUCRU) Indonesia)
Contact details available upon request